



MBARARA UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATICS

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

FLEET DATABASE MANAGEMENT SYSTEM

Standard Project Report for the Oil and Gas Sector

Course Unit	CSC1204 Database Systems
Assignment	Project 1
Student Name	Ntambi Sharif
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Lecturer	Keneth Baguma
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1. Executive Summary

The Fleet Database Management System is a centralized relational database solution designed to support fleet operations in the oil and gas sector. It manages vehicles, drivers, trips, routes, fuel usage, maintenance records, cargo movement, permits, inspections, and incident reporting.

The system addresses common operational challenges such as poor vehicle tracking, fuel wastage, delayed deliveries, weak driver accountability, and fragmented manual records. By storing operational data in one structured database, the organization can improve traceability, reporting, cost control, and decision-making.

2. Business Case

Oil and gas fleet operations involve fuel tankers, service vehicles, inspection vans, maintenance pickups, drivers, routes, depots, and cargo. Manual management of these resources creates delays, duplication, data loss, and limited visibility into operational performance.

2.1 Business Objectives

- Reduce manual record-keeping by storing fleet data in a centralized database.
- Improve vehicle tracking and utilization through structured trip monitoring.
- Enhance fuel management by tracking consumption, costs, and related vehicle activity.
- Strengthen maintenance planning through service records and next-service dates.
- Improve driver accountability by linking drivers to vehicles, trips, and incidents.
- Support route planning for efficient fuel distribution and logistics.
- Provide timely reports for management decisions.

3. System Overview

Fleet Managers	Supervise vehicles, routes, assignments, and operational performance.
Drivers	Record trip, fuel, route, and vehicle activity.
Technicians	Update service, repair, and maintenance records.
Management	Review reports and make operational decisions.
Administrators	Maintain users, master data, and system security.

3.1 Key Functions

- Stores vehicle details including type, registration number, condition, status, and depot.
- Manages driver profiles, license numbers, license expiry dates, and work status.
- Records trips, routes, distances, start times, end times, and trip status.
- Tracks fuel logs, maintenance activities, permits, cargo, inspections, and incidents.
- Produces structured data for operational reporting and accountability.

4. Benefits and Costs

Tangible Benefits	Reduced fuel theft, lower maintenance costs, faster record retrieval, improved vehicle use, and reduced idle time.
Intangible Benefits	Better planning, stronger accountability, improved safety, and increased client trust through reliable operations.
Tangible Costs	Computers, GPS devices, software development or purchase, training, maintenance, and updates.
Intangible Costs	Time required for users to learn the system and adjust to new workflows.

5. Risk Assessment

Technical	System crashes, data loss, and security threats.	Use backups, access controls, secure passwords, and regular maintenance.
Organizational	Resistance to change and limited management support.	Provide training, user involvement, and clear implementation leadership.
Operational	Wrong data entry, GPS errors, and delayed updates.	Use validation rules, supervision, and regular data reviews.
User Adoption	Some users may lack technical skills.	Provide practical training and simple user guides.

6. Database Design

The database uses related tables to represent fleet assets, people, routes, operations, safety, compliance, and cargo movement. Primary keys uniquely identify each record, while foreign keys connect related records across tables.

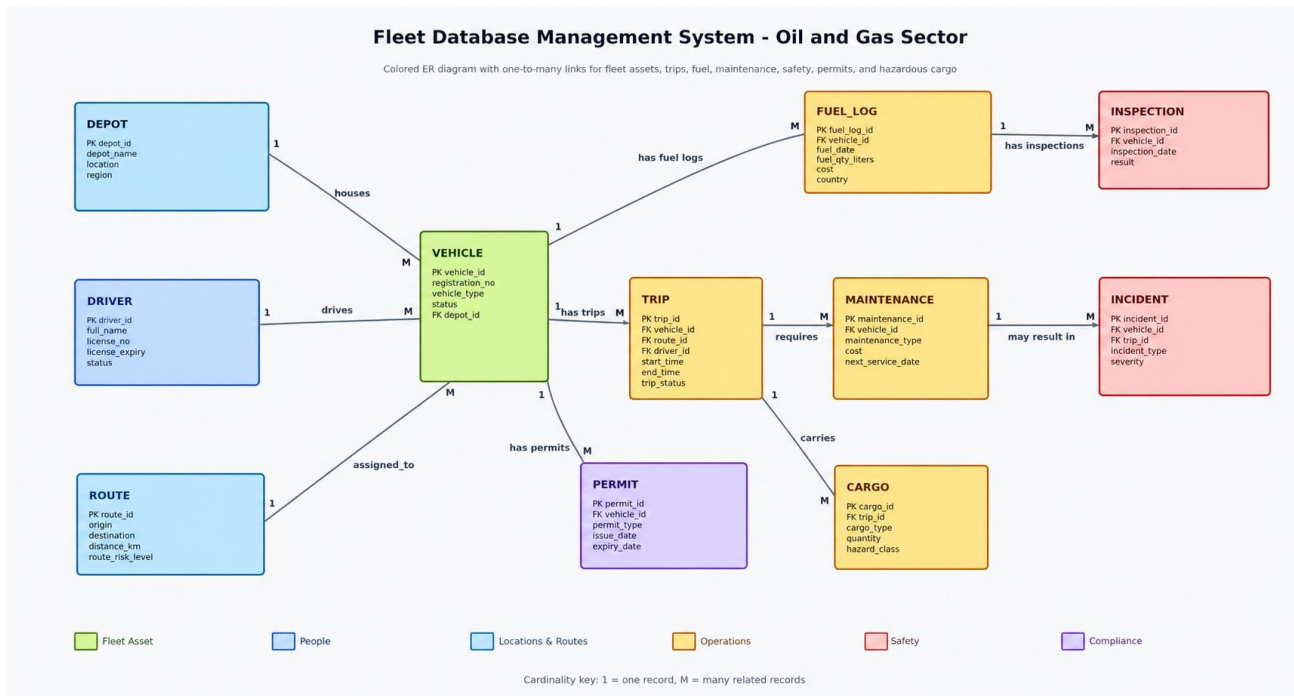


Figure 1: Entity relationship diagram for the Fleet Database Management System.

6.1 Main Entities

Depot	Stores depot names, locations, and regions.
Vehicle	Stores fleet vehicle registration, type, status, and depot assignment.
Driver	Stores driver identity, license details, and availability status.
Route	Stores origin, destination, distance, and route risk level.
Trip	Links vehicles, drivers, and routes for each transport activity.
Fuel_Log	Tracks fuel date, quantity, cost, and related vehicle activity.
Maintenance	Records maintenance type, cost, and next service date.
Permit	Stores vehicle permit details and expiry dates.
Cargo	Records cargo type, quantity, and hazard class for each trip.
Inspection and Incident	Support safety monitoring, inspection results, and incident reporting.

6.2 Relationship Summary

Depot to Vehicle	1:M	One depot can house many vehicles.
Driver to Vehicle	1:M	A driver can be assigned to vehicle operations.
Vehicle to Trip	1:M	One vehicle can make many trips.
Route to Vehicle	1:M	Routes are assigned to vehicles through operations.
Trip to Cargo	1:M	One trip can carry many cargo records.
Vehicle to Fuel_Log	1:M	One vehicle can have many fuel log records.
Vehicle to Maintenance	1:M	One vehicle can have many maintenance records.

7. Implementation Evidence

The following screenshots show sample database structure and records from the MariaDB implementation of the fleet_oil_gas_db database.

```
MariaDB [fleet_oil_gas_db]> describe driver;
```

Field	Type	Null	Key	Default	Extra
driver_id	int(11)	NO	PRI	NULL	auto_increment
full_name	varchar(120)	NO		NULL	
license_no	varchar(50)	NO	UNI	NULL	
license_expiry	date	NO		NULL	
status	enum('ACTIVE', 'SUSPENDED', 'ON_LEAVE', 'INACTIVE')	NO		ACTIVE	

5 rows in set (0.050 sec)

```
MariaDB [fleet_oil_gas_db]> describe fuel_log;
```

Field	Type	Null	Key	Default	Extra
fuel_log_id	int(11)	NO	PRI	NULL	auto_increment
vehicle_id	int(11)	NO	MUL	NULL	
driver_id	int(11)	NO	MUL	NULL	
fuel_date	date	NO		NULL	
quantity_liters	decimal(10,2)	NO		NULL	
cost	decimal(12,2)	NO		NULL	

6 rows in set (0.052 sec)

```
MariaDB [fleet_oil_gas_db]> describe incident;
```

Field	Type	Null	Key	Default	Extra
incident_id	int(11)	NO	PRI	NULL	auto_increment
vehicle_id	int(11)	NO	MUL	NULL	
driver_id	int(11)	NO	MUL	NULL	
trip_id	int(11)	NO	MUL	NULL	
incident_type	varchar(100)	NO		NULL	
severity	enum('LOW', 'MEDIUM', 'HIGH', 'CRITICAL')	NO		NULL	

6 rows in set (0.063 sec)

Figure 2: Sample SELECT queries for drivers, vehicles, maintenance, and depots.

```

MariaDB [fleet_oil_gas_db]> select*from driver;
+-----+-----+-----+-----+-----+
| driver_id | full_name | license_no | license_expiry | status |
+-----+-----+-----+-----+-----+
| 1 | John Mugisha | UG-DL-1001 | 2027-08-15 | ACTIVE |
| 2 | Sarah Nambi | UG-DL-1002 | 2028-02-20 | ACTIVE |
| 3 | Peter Okello | UG-DL-1003 | 2026-11-10 | ON_LEAVE |
| 4 | Grace Achieng | UG-DL-1004 | 2029-04-05 | ACTIVE |
+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [fleet_oil_gas_db]> select*from vehicle;
+-----+-----+-----+-----+-----+
| vehicle_id | registration_no | vehicle_type | status | depot_id |
+-----+-----+-----+-----+-----+
| 1 | UGX-245T | Fuel Tanker Truck | ASSIGNED | 1 |
| 2 | UAZ-918P | Pipeline Maintenance Pickup | AVAILABLE | 2 |
| 3 | KCN-774L | Crude Oil Tanker | ASSIGNED | 3 |
| 4 | UBH-502Q | Safety Inspection Van | MAINTENANCE | 1 |
+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [fleet_oil_gas_db]> select*from maintenance;
+-----+-----+-----+-----+-----+
| maintenance_id | vehicle_id | maintenance_type | cost | next_service_date |
+-----+-----+-----+-----+-----+
| 1 | 4 | Brake System Repair | 1250000.00 | 2026-06-10 |
| 2 | 1 | Routine Oil Service | 450000.00 | 2026-07-01 |
| 3 | 2 | Tyre Replacement | 980000.00 | 2026-08-15 |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

```

Figure 3: Sample table descriptions for driver, fuel_log, and incident entities.

8. Limitations

- The system requires reliable network access for timely updates.
- Initial setup costs may be high because of computers, GPS equipment, and training.
- The quality of reports depends on accurate and timely data entry.
- The system requires regular maintenance, backups, and security monitoring.

9. Conclusion

The Fleet Database Management System improves the efficiency, accountability, and safety of oil and gas fleet operations. By replacing fragmented manual records with a centralized relational database, the organization can control costs, monitor assets, improve driver accountability, and generate reliable reports for decision-making.

Appendix A: Additional Query Evidence

```

MariaDB [fleet_oil_gas_db]> select*from driver;
+-----+-----+-----+-----+-----+
| driver_id | full_name | license_no | license_expiry | status |
+-----+-----+-----+-----+-----+
| 1 | John Mugisha | UG-DL-1001 | 2027-08-15 | ACTIVE |
| 2 | Sarah Nambi | UG-DL-1002 | 2028-02-20 | ACTIVE |
| 3 | Peter Okello | UG-DL-1003 | 2026-11-10 | ON_LEAVE |
| 4 | Grace Achieng | UG-DL-1004 | 2029-04-05 | ACTIVE |
+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [fleet_oil_gas_db]> select*from vehicle;
+-----+-----+-----+-----+-----+
| vehicle_id | registration_no | vehicle_type | status | depot_id |
+-----+-----+-----+-----+-----+
| 1 | UGX-245T | Fuel Tanker Truck | ASSIGNED | 1 |
| 2 | UAZ-918P | Pipeline Maintenance Pickup | AVAILABLE | 2 |
| 3 | KCN-774L | Crude Oil Tanker | ASSIGNED | 3 |
| 4 | UBH-502Q | Safety Inspection Van | MAINTENANCE | 1 |
+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [fleet_oil_gas_db]> select*from maintenance;
+-----+-----+-----+-----+-----+
| maintenance_id | vehicle_id | maintenance_type | cost | next_service_date |
+-----+-----+-----+-----+-----+
| 1 | 4 | Brake System Repair | 1250000.00 | 2026-06-10 |
| 2 | 1 | Routine Oil Service | 450000.00 | 2026-07-01 |
| 3 | 2 | Tyre Replacement | 980000.00 | 2026-08-15 |
+-----+-----+-----+-----+-----+
3 rows in set (0.001 sec)

```

Figure A1: Database schema evidence from MariaDB.